

Week 4: JavaScript Basics + ES6

1. JS Syntax, Data Types, Variables (let/const/var)

JavaScript is a programming language used to make web pages interactive and dynamic. Learning JavaScript syntax is important because it defines how code is written and executed. Data types such as strings, numbers, booleans, arrays, and objects help store and manage different kinds of information. Variables like `let`, `const`, and `var` are used to store values in programs. Understanding these basics is essential for writing clean and functional JavaScript code.

2. Functions, Loops, Conditionals, Arrays, Objects

Functions are reusable blocks of code that perform specific tasks. Loops such as `for` and `while` are used to repeat tasks automatically. Conditionals like `if`, `else`, and `switch` help make decisions in a program. Arrays are used to store multiple values in a single variable, while objects store related data in key-value pairs. These concepts are fundamental for building logical and interactive applications.

3. Arrow Functions, Template Literals, Destructuring

ES6 introduced arrow functions, which provide a shorter and cleaner syntax for writing functions. Template literals allow easier string formatting and support embedding variables directly inside text. Destructuring is used to extract values from arrays and objects quickly and efficiently. These features improve readability and reduce complexity in JavaScript code.

4. Spread/Rest Operators, Modules

Spread and rest operators are useful for handling arrays, objects, and function parameters. The spread operator helps copy or merge data, while the rest operator collects multiple values into a single variable. Modules allow developers to split code into separate reusable files, making applications more organized and easier to maintain. These ES6 features are important in modern JavaScript development.

Conclusion

This week focused on JavaScript basics and ES6 features. By understanding syntax, variables, loops, functions, arrays, objects, and modern ES6 concepts, a strong foundation was built for creating dynamic and scalable web applications.